

NAVEEN S DAS

M.Tech, Quantum Technology · Indian Institute of Technology, Jodhpur
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SUMMARY

M.Tech scholar in **Quantum Technology** at IIT Jodhpur, working across **quantum error correction, quantum machine learning and quantum optimization**, with a prior career in data science and production ML. Comfortable moving between QEC decoding pipelines, variational quantum circuits, QUBO/Ising formulations and agentic AI systems, and turning research into working software. Deeply interested in where quantum computing, AI/ML and **cognitive science / neuroscience** meet — how intelligence is represented, and what quantum and learning machines can say about it.

EDUCATION

Degree / Certificate	Institute / Board	CGPA / %	Year
M.Tech, Quantum Technology	Indian Institute of Technology, Jodhpur	9.25	2026
PGP, Data Science	Great Lakes Institute of Management, Gurgaon	—	2022
B.Tech, ECE	Model Engineering College, Kochi	6.59	2020
Senior Secondary	Vidyodaya School, Thevakkal	91%	2016
Secondary	Campion School, Kochi	9.4	2014

EXPERIENCE

Infosys, Bangalore — Quantum Research Intern, Quantum Machine Learning May – July 2026

- Researched **quantum-friendly features and datasets** for QML — classical hardness, data-geometry complexity, kernel spectra and trainability — to identify where quantum models offer genuine utility.
- Built a **decision engine** routing incoming problems and datasets to classical, quantum or hybrid execution. **Patent application in progress.**
- Built **Quantum Agents** — agentic-AI-orchestrated quantum solvers for combinatorial optimization — delivering the full stack and a working prototype.
- Authored enterprise-grade product designs for quantum application software with agentic AI workflows for client deployments.

Gadgeon Smart Systems Pvt Limited, Kerala — Data Scientist 2022 – 2023

- Built and deployed **production-grade ML pipelines** for predictive maintenance and industrial analytics, enabling real-time fault detection in IoT-enabled manufacturing environments.
- Developed end-to-end predictive analytics systems (LSTM, Random Forest, XGBoost) on high-frequency sensor streams, reducing unplanned downtime and improving operational efficiency.
- Designed scalable preprocessing and feature engineering pipelines robust to noisy real-time data; mentored junior engineers in ML best practices.

PROJECTS

Neural Decoders for Quantum LDPC Codes — M.Tech Thesis Ongoing

Python · PyTorch · BP-OSD · Bivariate Bicycle Codes · Quantum Error Correction

- Studying whether coset-graded training objectives drive out-of-distribution size generalization in neural decoders for QLDPC codes, across code distance, noise regime and code family.
- Built the feasibility pipeline: constructed the $[[72,12,6]]$ bivariate bicycle code, sampled circuit-level syndromes, ran a BP-OSD baseline and quantified decoder degeneracy.

Q-FLUX — Quantum-Optimised Fleet Logistics with Live Digital Twin 2026

D-Wave Ocean (dimod, neal, Leap) · QUBO/Ising · NetworkX · OR-Tools CP-SAT · FastAPI · [repo](#)

- Formulated truck-to-order assignment as a **QUBO** — capacity, hazmat and assignment constraints as penalties; mileage, tolls, travel time and deadline lateness as cost — solved by annealing with a D-Wave Leap path.
- Built a live digital twin with disruption propagation (breakdowns, road closures, urgent orders) that isolates affected orders and re-optimises only that zone.
- Benchmarked the annealer against greedy and OR-Tools CP-SAT on identical objectives: **1.29% gap to proven optimum**, 8.13% saving versus riding out a disruption.

Brain-Computer Interface (BCI) — Canine Olfaction EEG Signal Classification

Python · MNE · Biosignal Processing · Neural Networks · Time-Series Analysis

- Built an end-to-end pipeline for EEG acquisition, preprocessing, feature extraction and neural-network classification.
- Applied time-frequency analysis and data storytelling for medical-grade signal interpretation.

Hybrid Quantum-Classical Neural Network for Image Classification

Ongoing

PennyLane · PyTorch · Qiskit · ResNet · Vision Transformers · Variational Quantum Circuits

- Designed a hybrid architecture embedding variational quantum circuits within classical deep learning backbones (ResNet / Vision Transformers).
- Optimised **quantum feature maps and ansatz design** to maximise expressive power within the qubit and depth constraints of NISQ processors.

BB84 Quantum Key Distribution Simulation with Decoy States

Python · Qiskit · Quantum Communication · QBER Analysis · Interactive Dashboard

- Simulated BB84 QKD with decoy-state extension; computed secure key rates and QBER under realistic channel noise, with a dashboard for real-time protocol analysis.

Retrieval-Augmented Generation (RAG) Application

GPT · Llama 3 · LangChain · Chroma & Pinecone · FastAPI · Docker

- Architected and deployed a production-ready RAG system with vector-database retrieval, reducing hallucination through structured prompt engineering; containerised with Docker and served via FastAPI.

SKILLS

Quantum Technology Quantum Error Correction (stabiliser & QLDPC codes, BP-OSD and neural decoders, circuit-level noise) · Quantum Machine Learning (variational circuits, quantum feature maps and kernels, hybrid models, trainability) · Quantum Optimization (QAOA, VQE, QUBO/Ising, quantum annealing) · Quantum Cryptography (BB84 / QKD) · NISQ noise mitigation

AI / ML PyTorch · TensorFlow · Scikit-Learn · Computer Vision (CNNs, ViTs) · NLP (LLMs, RAG) · LangChain / LangGraph · Agentic AI · Vector Databases · Model Deployment & Optimization

Quantum SDKs Qiskit · PennyLane · D-Wave Ocean SDK · OR-Tools · NetworkX · NISQ hardware

Programming & Data Python · C++ · SQL · NumPy · SciPy · Tableau · Data Analysis

MLOps & Cloud Git · Docker · CI/CD · FastAPI · Streamlit · MLflow · AWS · GCP

Key Courses Quantum Computation · Quantum Algorithms · Quantum Error Correction · Quantum Machine Learning · Quantum Optimization · Quantum Cryptography · Classical & Quantum Algorithms · Device Independent Quantum Technology

CERTIFICATIONS & ACHIEVEMENTS

- Smart India Hackathon (SIH) 2025 — Finalist · GATE 2025 Qualified (Data Science & AI)
- IBM Quantum Badges · Computational Neuroscience, University of Washington (Coursera)
- BizQuiz, IIT Jodhpur Techfest — 2nd Place · College and University level Football player